

Subject Description Form

Subject Code	COMP5574
Subject Title	Computational Economics and Algorithms
Credit Value	3
Level	5
Pre-requisite/ Co-requisite/ Exclusion	Nil
Objectives	The objectives of this subject are to: 1. Provide students with a foundational understanding of the game-theoretic issues behind systems involving computation such as online networks. 2. Enable students to learn how algorithms and algorithmic thinking can help with designing better decision and allocation mechanisms in the offline world.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: a. understand the basic concepts of game theory and mechanism design; b. apply computational methods to solve optimization problems in economics, and design algorithms that incorporate economic principles for applications such as auctions, pricing, and resource allocation; c. analyse the impact of information technology on market structures and economic strategies, and demonstrate the ability of critical reading and analysis through paper reviews and panel discussions, enhancing abilities to write reviews and analyse papers from multiple perspectives; d. evaluate the role of data and computational analysis in economic decision-making and policy formulation; e. explore the ethical implications of computational economics in the context of privacy, fairness, and market regulation.
Subject Synopsis/ Indicative Syllabus	1. Introduction to Computational Economics • Overview of the field • Intersection of economics and computer science 2. Game Theory and Strategic Interaction • Basic concepts of game theory • Nash equilibrium

	• Price of Anarchy • Repeated games and evolutionary stability
	3. Mechanism Design and Auction Theory • Designing mechanisms for desired outcomes • Auction formats and bidding strategies • Revenue equivalence and incentive compatibility
	4. Market Design and Matching Algorithms • Two-sided matching markets • Stable matching and the Gale-Shapley algorithm • Market design in practice: labour markets, school choice, organ donation
	5. Optimisation and Linear Programming • Linear programming in economic modelling • Duality and sensitivity analysis • Network flows and integer programming
	6. Computational Models of Economic Dynamics • Agent-based modelling • Computational general equilibrium models • Dynamic stochastic models • Voting systems, cake cutting, resource allocation and their computational aspects
	7. Data Analysis and Econometrics • Big data in economics • Machine learning techniques for economic data • Causal inference and experimental design
	8. Digital Markets and Platforms • Economics of digital goods and services • Platform competition and network effects • Pricing strategies and revenue management • Cryptocurrencies and Bitcoin
	9. Algorithmic Game Theory and the Internet • The role of algorithms in online markets • Search engines and online advertising • Social networks and information diffusion
	10. Ethical Considerations in Computational Economics • Privacy and data protection • Algorithmic bias and fairness • Regulation of digital markets

Teaching/Learning Methodology	Lectures provide students with the main concepts of economics and computation, together with comprehensive examples for easy understanding. Students will engage in hands-on exercises to apply computational techniques to economic problems. Presentations will encourage collaboration and practical application of course material. A written assignment will be utilised in the course to help students develop analysis and design skills.																																													
Assessment Methods in Alignment with Intended Learning Outcomes	<table><tr><th rowspan="2">Specific assessment methods/tasks</th><th rowspan="2">% weighting</th><th colspan="5">Intended subject learning outcomes to be assessed (Please tick as appropriate)</th></tr><tr><th>a</th><th>b</th><th>c</th><th>d</th><th>e</th></tr><tr><td>1. Assignment and presentation</td><td>10%</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td></tr><tr><td>2. Quiz</td><td>20%</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td></tr><tr><td>3. Exam</td><td>70%</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td></tr><tr><td>Total</td><td>100 %</td><td colspan="5"></td></tr></table> Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes: The assignment and quiz assesses students' comprehension of the fundamental concepts in economics and computation. The presentation evaluates students' critical and creative thinking capabilities, problem-solving skills, and their ability to apply the techniques and tools they have learnt to real-world applications. The exam is used to assess independent problem-solving and critical thinking.						Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)					a	b	c	d	e	1. Assignment and presentation	10%	✓	✓	✓	✓	✓	2. Quiz	20%	✓	✓	✓	✓	✓	3. Exam	70%	✓	✓	✓	✓	✓	Total	100 %					
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Student Study Effort Expected	Class contact:																																													
	▪ Lecture/Tutorial/Lab/Paper panel				39 Hrs.																																									
	Other student study effort																																													
	▪ Assignment, preparing presentation, reviewing and preparing for quiz and exam, reading papers, self-study				81 Hrs.																																									
	Total student study effort				120 Hrs.																																									
Reading List and References	1. T. Roughgarden. Twenty Lectures on Algorithmic Game Theory, Cambridge University Press, 2016																																													

	2. N. Nisan, T. Roughgarden, E. Tardos and V. Vazirani, Algorithmic Game Theory, 2007 3. Felix Brandt, Vincent Conitzer, Ulle Endriss, Jérôme Lang and Ariel D. Procaccia. Handbook of Computational Social Choice, Cambridge University Press, 2016 4. Martin J. Osborne and Ariel Rubinstein, A Course in Game Theory, MIT Press, 1995 5. Andreu Mas-Colell, Michael D. Whinston, Jerry R. Green. Microeconomic Theory, Oxford University Press, 1995 6. David Easley and Jon Kleinberg. Networks, Crowds, and Markets: Reasoning About a Highly Connected World, Cambridge University Press 2010
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